

**SRI MANAKULA VINAYAGAR ENGINEERING COLLEGE**

(Approved by AICTE, New Delhi & Affiliated to Pondicherry University)  
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Madagadipet, Puducherry - 605 107

**UNIT: I****1. Define Doping?**

Adding of impurities to the intrinsic semiconductor in order to conduct electricity is known as doping.

**2. What is Biasing?**

Biasing is a process of applying external DC voltage to the device so that it can operate in a desired way.

**3. What is Diffusion capacitance?**

It is the time delay in moving charges across the junction by diffusion process. When the junction is forward biased it takes place.

**4. Define Ripple factor?**

It is the measure of AC components present in the DC output.

$$\gamma = \sqrt{[(I_{rms}/I_{dc})^2 - 1]}$$

Ripple factor is defined as the ratio of R.M.S. value of AC component to DC component.

**5. Define Ionization?**

Ionization is a process of converting electrically neutral atoms into charged ions.

**6. Define V-I Characteristics?**

The response of the diode when connected in an electrical circuit can be judged from its characteristics known as volt- ampere characteristics commonly known as V-I characteristics.

**7. Define Voltage Regulator?**

It is the measure of the circuit's ability to maintain a constant output voltage even when either input voltage or load current varies.

**8. What are the types of voltage regulators?**

\* Zener diode shunt regulator

\* Transistor shunt regulator

\*Transistor series regulator

\*Controlled transistor series regulator \*Monolithic regulators

### **9. What is the need for filters in rectifiers?**

Filters are used to eliminate the unwanted part of the signal from the available signal. Filter components consists of inductor and capacitor.

### **10. Define Clippers?**

Clipper circuits are defined as circuit which removes the unwanted part of a signal from the available signal without destroying the rest of the signal. Clippers are also called as limiters or slicers.

### **11. What is a Series Clipper?**

In a clipper circuit if a diode is connected in series with the load then it is termed to be series clipper.

### **12. What is a Parallel or shunt Clipper?**

If a diode is connected parallel to the load it is called parallel clipper.

### **13. Define Clamper?**

Clamper circuit is used to add DC level to the AC output signal. It consists of capacitor, diode and load resistor.

### **14. Define Voltage Multiplier?**

Voltage multipliers are used to step up the output voltage to twice or thrice more than the peak voltage of the input.

### **15. What is voltage doubler?**

Voltage doubler is defined as the circuit which gives voltage equal to twice the amount of the input.

### **16. What is voltage Tripler?**

Voltage Tripler circuit is derived from the half wave doubler circuit. It produces output three times than that of the inputs.

### **17. What are the types of Semiconductor?**

1. Intrinsic semiconductor
2. Extrinsic semiconductor.

**18. What is extrinsic semiconductor?**

If certain amount of impurity atom is added to intrinsic semiconductor the resulting semiconductor is Extrinsic or impure Semiconductor.

**19. What is intrinsic semiconductor?**

It is the purest form of semiconductor. These are perfectly insulator. A type of semiconductor that has a low level of electron movement at any temperature above 0°C (32°F). Silicon is an intrinsic semiconductor.

**20. Define depletion layer?**

A wall of positively immobile charges on p-side and a wall of negatively immobile charges on n-side form a region called depletion region. In depletion region there is no mobile charge carrier. It acts as a barrier.

**21. What is potential barrier?**

The electric field produced due to collision of opposite charges on depletion layer is known as potential barrier.

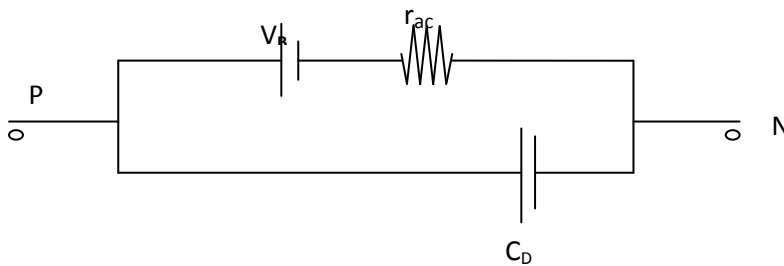
**22. What are the types of filters?**

1. Capacitor filter
2. Inductor filter
3. Choke input filter (LC filter)
4. Capacitor input filter ( $\pi$  filter)

**23. Define rectifiers and its types?**

A rectifier converts an A.C Voltage to pulsating D.C. The various types of rectifiers are:

- 1: half wave rectifier
- 2: full wave rectifier
- 3: bridge rectifier

**24. Equivalent circuit for p-n junction diode?**

$V_B$  -> Diffusion capacitance,  $r_{ac}$  -> AC resistance,  $C_D$  -> Opposing battery

**25. What is break down? What are its types?**

When the reverse voltage across the pn junction is increased rapidly at a voltage the junction breaks down leading to a current flow across the device. This phenomenon is called as break down and the voltage is break down voltage. The types of break down are

- i) zener break down.
- ii) Avalanche breakdown.

**26. What are the junctions in transistor?**

- i) Base Emitter junction,
- ii) collector Base junction.

**27. List some applications of zener diode?**

Zener diode finds wide commercial and industrial applications. Some of their common Applications are:

- ✓ As voltage regulators.
- ✓ As peak clippers or voltage limiters.
- ✓ For wave shaping.
- ✓ for meter protection against damage from accidental application of excessive voltage.
- ✓ As a fixed reference voltage in a network for biasing and comparison purposes and for calibrating voltmeters.

**28. What is the function of a positive clamper?**

It shifts the signal towards the positive side such that the negative side of the signal reduces to zero.

**29. What is the function of a negative clamper?**

It shifts the signal towards to negative side such that the positive side of the signal reduces to zero.

**30. List the applications of Clampers?**

- ✓ They are used in T.V. receivers to restore the original DC reference signal (corresponding to the brightness level of the picture) to the video signal.
- ✓ They are used to produce a DC voltage is a multiple of peak AC input voltage i.e., they are used as voltage multipliers.
- ✓ They are used to supply power to high voltage/low current devices like CRTs used in T.V receivers, CROs and computer displays.

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**UNIT-II****1. Define transistor?**

Transistor is transfer resistor. When a third doped element is added to a diode in such a way that the PN junctions are formed. The resulting device is known as transistor.

**2. Define BJT?**

Bipolar junction transistor is often applied to 3-terminal devices. Bipolar reflects that holes and electrons participate in the injection process into the oppositely polarized material. If only one carrier is employed (i.e) electron or hole it is called as unipolar. Eg: schottky diode.

**3. What are the types of transistor?**

There are two types of transistor namely,

- NPN transistor
- PNP transistor

**4. Define Emitter?**

The section on one side that supplies charge carriers( $e^-$  or holes) is called emitter. The emitter is always forward bias with respect to base so that it can supply a large number of majority carriers (i.e)holes if emitter is p-type and  $e^-$  if that emitter is n-type.

**5. Define Base?**

The middle section which forms two pn-junction between the emitter and the collector is called base. The base-emitter junction is forward bias, allowing low resistance for the emitter circuit. The base should be much thinner than the emitter and it is lightly doped.

**6. Define Collector?**

The section on the other side that collects the charges is called collector. The collector is always reverse biased. Its function is to remove charges from its junction with the base.

**7. Define Emitter diode?**

The transistor has two PN-junctions (i.e) it is like two diodes. The junction between emitter and base may be called emitter -base diode or emitter diode.

**8. Define amplification factor?**

The ratio of change in output current to change in input current is known as amplification factor.

$$\text{Amplification factor} = d I_O / d I_{IN}$$

**9. What are the various transistor configurations?**

The transistor can be classified into three types of configuration .They are

- common base configuration
- common emitter configuration
- common collector configuration

**10. Compare transistor configuration?**

| SL.NO. | CHARACTER         | CB                             | CE                    | CC                 |
|--------|-------------------|--------------------------------|-----------------------|--------------------|
| 1.     | Input resistance  | Low                            | Low                   | very high          |
| 2.     | Output resistance | High                           | Very high             | low                |
| 3.     | Voltage gain      | 150                            | 500                   | less than 1        |
| 4.     | application       | For high frequency application | For audio application | impedance matching |

**11. What is transistor biasing and what is the need for it?**

The proper flow of zero signal collector current and the maintenance of proper collector emitter voltage during the passage of signal is known as transistor biasing. The need for this is to

keep the base – emitter junction properly forward and collector-base junction properly reverse biased.

### 12. Define stability factor?

Is defined as the rate of change of collector current  $I_c$  with respect to the collector leakage current  $I_{co}$  at constant base current and beta is called stability factor  $S = dI_c / dI_{co}$

### 13. Define faithful amplification?

The process of raising the strength of a weak signal without any change in its general shape is known as faithful amplification.

### 14. Define transistor biasing types?

- Fixed bias
- Collector to base bias
- Self bias

### 15. Discuss the merits and demerits of various transistor biasing methods?

#### i) Fixed bias:

##### Merits:

- This biasing ckt is very simple as only one resistor is used
- Biasing condition can be easily be set and the calculations are simple

##### Demerits:

- This method provides poor stabilization
- Stability factor is very high.

#### ii) Collector to base bias:

##### Merits:

- It is a simple method as it requires only one resistance

##### Demerits:

- This circuit provides a negative feedback which reduces the gain
- The circuit does not provide good stabilization because  $S$  is fairly high though it is lesser than that of fixed bias

#### iii) Voltage divider bias:



**Merits:**

- Most widely used method
- The smallest possible value of S is obtained

**16. Define load line?**

The straight line that is used for the analysis of collector current for various collector emitter voltages is called as load line.

**17. Define q-point?**

The zero signal values of  $I_c$  and  $V_{ce}$  are known as operating point. It is also called as quiescent (silent) point. The transistor is silent at this point where there is no input signal.

**18. Define voltage gain?**

It is defined as the ratio of output voltage to input voltage

$$A_v = V_o / V_{in}$$

**19. Define current gain?**

It is defined as the ratio of output current to input current.

$$A_i = I_o / I_{in}$$

**20. Define input impedance?**

It is defined as ratio of input voltage to input current.

$$Z_i = V_i / I_i$$

**21. Define output admittance?**

It is defined as the ratio of output current to output voltage.

$$Y_o = I_o / V_o$$

**22. Define power gain?**

It is defined as the ratio of load power to source power.

$$A_p = P_l / P_i$$

**23. What is h-parameters?**

To obtain the accurate results of i/p and o/p ckts four parameters of transistors are used. These parameters are called as hybrid (or)h-parameters of the transistor.

**24. Define FET?**

Field effect transistor has large input impedance. It is generally much less noisy than BJT.

**25. Discuss the types of FET?****1. JFET**

- n-channel JFET
- p-channel JFET

**2. MESFET (Metal Semiconductor FET)**

- n-channel MESFET
- p-channel MESFET

**3. MOSFET (Metal Oxide Silicon FET)**

- DE-MOSFET(Depletion MOSFET)
  - >n-channel DE-MOSFET
  - >p-channel DE-MOSFET
- E-MOSFET(Enhancement MOSFET)
  - >n-channel E-MOSFET
  - >p-channel E-MOSFET

**26. Define the terminals of JFET?**

The two PN junction forming diodes are connected internally and a common terminal called gate (G).The other terminals are source(S) and drain(D).

**27. Define the breakdown region in FET?**

If  $V_{ds}$  is increased beyond pinch off voltage ( $V_p$ ) the drain current  $I_d$  remains constant. If  $V_{ds}$  is further increased the drain current increases very rapidly and the device may be destroyed. This region is called breakdown region.

**28. Define pinch off voltage?**

The voltage  $V_{ds}$  at which the current  $I_d$  reaches to its constant saturation level is called pinch off voltage.

**29. What are the parameters of JFET?**

- Transconductance ( $g_m$ )

- Input resistance ( $r_i$ )
- Drain to source resistance ( $r_d$ )
- Amplification factor
- Power dissipation ( $p_d$ )

**30. Define Tran's conductance?**

It is defined as the change in drain current for the given change in gate to source voltage with drain to source voltage as constant.

$$g_m = d I_d / V_{gs}, \text{ at constant } V_{ds}$$

**31. Define amplification factor in FET?**

It is defined as the ratio of change in drain to source voltage to change in gate to source voltage.

$$\text{Amplification factor} = d V_{ds} / V_{gs}$$

**32. What are the types biasing in FET?**

- Fixed bias
- Voltage divider bias
- Self bias

**33. What are the FET amplifier configurations?**

- Common source amplifier
- Common drain amplifier
- Common gate amplifier

**34. List some applications of JFETs?**

- ✓ Used as buffers in measuring equipment, receivers and other general purpose devices.
- ✓ Used in RF amplifiers of FM tuners and communication equipment.
- ✓ Used in mixer circuits in FM and TV receivers and communication equipment.
- ✓ Used in cascade amplifiers in measuring and test equipment.
- ✓ Used as voltage variable resistor (VVR) in OP-AMPs and tone controls.
- ✓ Used in hearing aids and inductive transducers.
- ✓ Used in oscillator circuits.

- ✓ as the physical size is small; it finds use in digital circuits in computers, large Scale integration (LSI) and memory circuits.
- ✓ Used as current sources.

**35. Define Drain resistance?**

It is the ratio of change in Drain – source voltage ( $\Delta V_{DS}$ ) to the change in Drain current ( $\Delta I_D$ ) at constant gate source voltage ( $V_{GS}$ ).

$$R_D = \frac{\Delta V_{DS}}{\Delta I_D} \text{ at constant } V_{GS}.$$

**36. What are the limitations of h parameters?**

The h parameters has the following limitations,

- a. The accurate calculation of h parameters is difficult.
- b. A transistor behaves as a two port network for small signals only, hence h parameters can be used to analyze only the small signal amplifiers.



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### UNIT-III

#### 1. Define large signal amplifier.

Power amplifiers are also called as large signals amplifiers. These amplifiers use a large part of their a.c. load line for operation. These amplifiers have high efficiency and operate in non-linear region.

#### 2. Define power amplifier.

It is an amplifier which is capable of providing a large amount of power to the load. The a.c power developed at the output of a power amplifier is controlled by the input signal. Power amplifiers are also known as 'audio amplifier'.

#### 3. Define voltage amplifier.

Voltage amplifier is a small signal amplifier that produces an increased voltage across a high resistance load. Voltage amplifier uses only 10% of the load line for amplification.

#### 4. Compare voltage and power amplifier.

| Sl.no | Voltage amplifier                                                             | Power amplifier                                                                                            |
|-------|-------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|
| 1.    | Voltage amplifier is small signal amplifier.                                  | Power amplifier is large signal amplifier.                                                                 |
| 2.    | Voltage amplifier produces an increased voltage across a high resistance.     | Power amplifier produces an increased power or appreciable amount of current across a low resistance load. |
| 3.    | The value of beta is greater than 100.                                        | The value of beta is low, typically 5 to 20 ohm.                                                           |
| 4.    | Input resistance of the transistor is low when compared to output resistance. | Input resistance is high as compared to output resistance.                                                 |

|    |                                                                     |                                                                   |
|----|---------------------------------------------------------------------|-------------------------------------------------------------------|
| 5. | Power output is low.                                                | Power output is high.                                             |
| 6. | Voltage amplifier uses only 10% of the load line for amplification. | Power transistors use larger part of load line for amplification. |
| 7. | Output resistance is high.                                          | Output impedance is low.                                          |

**5. What are the parameters used in power amplifiers.**

The parameters that are used in power amplifiers are:

- Collector efficiency
- Distortion
- Power dissipation capability

**6. Define collector efficiency.**

Collector efficiency is defined as the ratio of a.c. output power to the zero signal power supplied by the battery of a power amplifier.

**7. Define distortion.**

In class-B power amplifier, the transistor conducts only for one -half cycle and remains OFF during other half cycle. This type of output signal gives a large distortion.

**8. What are two types of power amplifiers?**

One of the ways of classifying amplifiers is based on the position of the 'quiescent point' on the load line. The types are:

- Class-A
- Class-B
- Class-AB
- Class-C

**9. Define class-A amplifier.**

If the collector current flows at all times during the full cycle of the input signal, the power amplifier is known as class-A power amplifier.

**10. Define class-B amplifier.**

If the collector current flows only during the positive half cycle of the input signal, then it is called as class-B power amplifier.

**11. Define class-AB amplifier.**

If the collector current flows for more than  $180^\circ$  but less than  $360^\circ$  of the input signal, then power amplifier is known as class-AB power amplifier.

Class AB amplifiers combine Class A and Class B to achieve an amplifier with more efficiency than Class A but with lower distortion than class B.

**12. Define class-C amplifier.**

If the collector current flows for less than half cycle of the input signal, it is called class-C power amplifier.

**13. Define push-pull concept.**

Two transistors are operated in class-B, in such a manner that one transistor conducts during positive half cycle of the input waveform and the other conducts during the negative half-cycle of the output current though the load is of full-cycle. This type of operation is called as push-pull.

**14. Why push –pull concept is required?**

In class-B power amplifier, the Q-point current is zero. Therefore the transistor conducts for only half of the cycle of the input waveform, so the output is not a faithful reproduction of the input. In order to overcome this problem, the push pull concept is required.

**15. What are the types of class-A power amplifiers?**

Class-A power amplifiers consists of two types:

- Direct coupled (or) series fed class-A power amplifier.
- Transformer coupled (or) single ended class-A power amplifier.

**16. What are the types of class-B power amplifiers?**

Class-B power amplifiers are of two types:

- Transformer Coupled class-B push-pull amplifier (or) Double ended class-B amplifier.
- Transformer-less class-B push-pull amplifier

- Complementary Symmetry class-B push-pull amplifier.

### 17. Define harmonic distortion.

Harmonic distortion is defined as the ratio of  $n^{\text{th}}$  frequency component's amplitude to the amplitude of the fundamental frequency.

$$\% n^{\text{th}} \text{ harmonic distortion} = \% D_n = (|A_n|/|A_1|) \times 100\%$$

### 18. Define voltage gain of class-A power amplifier.

Voltage gain of class-A power amplifier is defined as the ratio of the load resistance ( $r_L$ ) to the a.c emitter diode resistance ( $r_e$ ).

$$A_v = r_L / r_e$$

where  $r_L$  = A.c load resistance whose value is equal to parallel combination of the collector resistance ( $R_C$ ) and the load resistance ( $R_L$ ) i.e.,  $r_L = R_C \parallel R_L$

### 19. What are the characteristics of class-A power amplifiers?

- The output current flows during the entire cycle of a.c input signal.
- Overall efficiency :

It is defined as the ratio of the a.c power delivered to the load to the total power supplied by the d.c source. The overall efficiency is given by,

$$\eta_o = (\text{average a.c output power}) / (\text{average d.c input power})$$

- Collector efficiency:

It is defined as the ratio of the a.c power delivered to the load, to the power supplied by the d.c source to the transistor. The collector efficiency is given by,

$$\eta_c = (\text{average power delivered to load}) / (\text{average d.c input power to the transistor})$$

### 20. Discuss the efficiency of class-A power amplifier.

#### Overall efficiency:

It is defined as the ratio of the a.c power delivered to the load to the total power supplied by the d.c source. The overall efficiency is given by,

$$\eta_o = (\text{a.c power delivered to the load}) / (\text{total power supplied by d.c source})$$

$$\eta_o = 25\%$$



**Collector efficiency:**

It is defined as the ratio of the a.c power delivered to the load, to the power supplied by the d.c source to the transistor. The collector efficiency is given by,

$$\eta_c = (\text{average a.c output power})/(\text{average d.c input power to the transistor})$$

$$\eta_c = 50\%$$

**21. Significance of transformer in class-A and class-B amplifiers.****Class-A:**

In direct coupled class A amplifier, the load resistance is directly connected in the output circuit of the amplifier. So, the overall efficiency does not exceed 25%. This problem is solved by using a transformer for coupling the load to the amplifier stage.

**Class-B:**

A transistor in class-B power amplifier conducts current for only one half cycle of the input signal. To obtain the output for the full cycle of the signal, it is necessary to use two transistors and each of the opposite half cycles and the combined operation gives full cycle to output.

**22. Define step down turn ratio.**

If  $N_1 > N_2$ , the transformer reduces the voltages in proportion to the turns ratio ( $N_2/N_1$ ) and steps up the current in the same ratio. It is given by,

$$\left(\frac{V_1}{I_1}\right) = \left(\frac{1}{n^2}\right)\left(\frac{V_2}{I_2}\right)$$

Where  $n = N_2/N_1$

$V_1/I_1$  - load resistance (or) resistance at primary side.

$V_2/I_2$  - resistance ( $R_L$ ).

$$R'_L = (1/n^2)R_L$$

This equation is called step down turnS ratio.

**23. Discuss the efficiency of class-B power amplifier.**

Efficiency is defined as the ratio of the a.c power delivered to the load to the total power supplied by the d.c source. The circuit efficiency of class-B push pull amplifier is 78.5% which is higher than that of class-A whose value is 25%.

$$\eta_o = (\text{a.c power delivered to the load})/(\text{total power supplied by d.c source})$$

**24. Define phase splitter.**

The transformer coupled class-B power amplifier consists of two centre tapped transformer T1 and T2 and two identical transistors Q1 and Q2. The transformer T1 is the input transformer and is called as 'phase splitter' and is required to produce two signal voltages, which are  $180^\circ$  out of phase with each other.

**25. Define cross over distortion.**

The transistor does not conduct until the input signal voltage exceeds 0.7v for Si and 0.3v for Ge. Because of this, there is a time interval between the positive and negative alternatives of the input signal when neither of the transistor is conducting. The resulting distortion in the output signal is called as 'cross-over distortion'.

**26. What are the characteristics of class-B power amplifier?**

- The output current flows only for one half cycle of the input signal.
- The average current drawn by the circuit is smaller than that of class-A. So that the amount of power dissipated by the transistor is less in class-B.
- The overall  $\eta$  of the circuit is higher than that of class-A (i.e) about 78.5%.

**27. What are the advantages of class-B power amplifiers?**

- The circuit efficiency of a class-B push pull amplifier is 78.5% which is much higher than that of class-A whose value is 25%.
- The use of push pull system in the class-B amplifier eliminates even order harmonics in the a.c output signal.
- Because of absence of harmonics , the circuit gives more output.
- There is no d.c components in the output signal.

**28. How the cross over distortion can be eliminated?**

The cross over distortion can be avoided by applying a slight forward bias equal to 0.7v for Si and 0.3v for Ge to the base-emitter junction of both the transistors.

**29. Define amplifier distortion.**

The difference between the output and the input signal of an amplifier is called as amplifier distortion. The two types of amplifier distortion are namely amplitude distortion and cross over distortion.

**30. Define duty cycle.**

Duty cycle is defined as the ratio of ON period of the transistor to total period of the pulses. The width of current pulses represents ON period of transistor.

$W$  = width of pulse

$T$  = period of the pulses

The duty cycle  $D$  is given by,

$$D = W/T$$

**31. Define tank circuit.**

A circuit, which produces electrical oscillations of any desired frequency, is known as oscillatory circuit. This circuit consists of two reactive elements namely, inductor ( $L$ ) and a capacitor ( $C$ ) connected in parallel with each other and is known as LC circuit or Tank circuit.

The resonant frequency of the tank circuit is determined by  $f_r = 1/(2\pi\sqrt{LC})$ .

**30. Compare push pull class-B and complementary class-B power amplifiers.**

| Sl.no | Push pull class-B                                                   | Complementary symmetry Class-B                                      |
|-------|---------------------------------------------------------------------|---------------------------------------------------------------------|
| 1.    | Both the transistors are similar either pnp or npn.                 | Transistors are complementary type (i.e).one npn and other pnp.     |
| 2.    | The transformers are used to connect the load as well as the input. | The circuit has no transformer                                      |
| 3.    | Frequency response is poor.                                         | Frequency response is improved.                                     |
| 4.    | Due to transformers, the circuit is bulky, costly and heavier.      | The circuit is not bulky and costly because of lack of transformer. |
| 5.    | Dual power supply is not required.                                  | Dual power supply is required.                                      |
| 6.    | Efficiency is higher than class-A.                                  | The efficiency is higher than push pull Class B.                    |

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**UNIT-IV****1. What is feedback amplifier?**

The feedback is a process of injecting some energy from output and returns it back to the input. The amplifiers, which use the feedback principle, are called feedback amplifiers.

**2. Define positive feedback.**

If the feedback signal (voltage or current) applied is *in phase* with the input signal and thus increases it, then it is called as positive feedback (or) regenerative feedback (or) direct feedback.

**3. Define negative feedback.**

If the feedback signal (voltage or current) applied is *out of phase* with the input signal and thus decreases it, then it is called as negative feedback (or) degenerative feedback (or) inverse feedback.

**4. What is piezo-electric effect?**

When an a.c. voltage is applied across the quartz crystal, it vibrates at the frequency of the applied voltage. This effect is called piezoelectric effect. Conversely, if a mechanical force is applied to vibrate a quartz crystal, it generates an a.c voltage.

**5. List out advantages and disadvantages of negative feedback.**Advantage:

- Increased stability
- Increased bandwidth
- Less amplitude and harmonic distortion
- Decreased noise
- Less frequency distortion
- Less phase distortion
- Input and output resistances can be modified as desired

Disadvantage:

- It reduces gain of amplifier.

**6. Disadvantage of positive feedback.**

An amplifier with positive feedback cannot have a stable voltage gain, because its gain increases with feedback. The increased voltage gain will further increase the feedback. This process will continue till the feedback ( $\beta.A_v$ ) becomes equal to unity. Under this conditions, an amplifier starts oscillating.

**7. Write the classification of amplifier.**

Linear Amplifiers can be classified into four broad categories

- (i) Based on input
  - a) Small signal amplifier b) Large signal amplifier
- (ii) Based on output
  - a) Voltage amplifier b) Power amplifier
- (iii) Based on transistor configuration
  - a) Common emitter amplifier b) Common base amplifier
  - b) Common collector amplifier
- (iv) Based on biasing condition
  - a) Class-A b) Class-B c) Class-AB d) Class-C

**8. Define sensitivity.**

The term  $1/(1 + \beta.A_v)$  is called sensitivity. It indicates the ratio of percentage change in voltage gain with feedback to the percentage change in voltage gain without feedback.

**9. Define desensitivity.**

The reciprocal of the term sensitivity is called desensitivity.

$$\text{Desensitivity} = (1 + \beta.A_v)$$

Desensitivity indicates the factor by which the voltage gain has been reduced due to feedback.

**10. List out the four types of feedback that connecting the feedback signal from output to input.**

- Voltage series feedback connection
- Voltage shunt feedback connection

- Current series feedback connection
- Current shunt feedback connection

### 11. What is Gain stability?

The voltage gain of the feedback amplifier ( $A_v'$ ) depends on the feedback network. (i.e.  $A_v' = 1/\beta$ ). It means that the voltage gain of the feedback amplifier does not change with the variation in transistor parameters due to temperature, ageing, etc. Therefore, the use of negative feedback provides a greater stability to the voltage gain of the amplifier.

### 12. Define bandwidth.

Bandwidth is defined as the difference between upper cutoff frequency to the lower cutoff frequency.

$$BW = (\text{upper cutoff frequency}) - (\text{lower cutoff frequency}).$$

The negative feedback increases the bandwidth of an amplifier by a factor  $(1 + \beta A_v)$ . Thus, the bandwidth of amplifier with feedback ( $BW'$ ) is greater than bandwidth of amplifier without feedback ( $BW$ ).

$$BW' = (1 + \beta A_v) BW$$

### 13. Compare the negative feedback amplifier characteristics.

| Parameter         | Voltage series                                  | Current series                           | Current shunt                                   | Voltage shunt                                   |
|-------------------|-------------------------------------------------|------------------------------------------|-------------------------------------------------|-------------------------------------------------|
| Input resistance  | $R_i' = R_i(1 + \beta A_v)$<br>Increases        | $R_i' = R_i(1 + \beta A_v)$<br>Increases | $R_i' = \frac{R_i}{1 + \beta A_v}$<br>decreases | $R_i' = \frac{R_i}{1 + \beta A_v}$<br>Decreases |
| Output resistance | $R_o' = \frac{R_o}{1 + \beta A_v}$<br>decreases | $R_o' = R_o(1 + \beta A_v)$<br>increases | $R_o' = R_o(1 + \beta A_v)$<br>increases        | $R_o' = \frac{R_o}{1 + \beta A_v}$<br>decreases |

### 14. What is an oscillator?

An oscillator is a device, which produces an output signal of any desired frequency without the application of an external input signal. It keeps producing an output signal as long as

dc power is supplied. An oscillator does not require any external signal to start or maintain energy conversion process.

**15. What are the applications of oscillators?**

Oscillators are commonly used in

- i) Radio and television transmitters and receivers
- ii) In radar, closed loop control systems.
- iii) In Filters, Analog communication channels.

**16. What are classifications of oscillator?**

Depending on the nature of the generated waveform, oscillators may be classified into two categories

- i) Sinusoidal or harmonic oscillators
- ii) Non- Sinusoidal or Relaxation oscillators

**17. Define sinusoidal oscillator?**

An oscillator which provide an output having a sinusoidal waveform of a definite frequency are called sinusoidal or harmonic oscillator. Such oscillator can provide output at frequencies ranging from 20 Hz to 1 GHz.

**18. Define relaxation oscillator?**

The oscillators, which provide an output having a square, rectangular or sawtooth waveform, are called non-sinusoidal (or) relaxation oscillators. Such oscillator can provide output at frequencies ranging from 0 Hz to 20 MHz.

**19. Define loop gain.**

The product of the feedback ratio ( $\beta$ ) and the voltage gain ( $A_v$ ) of a feedback amplifier is called 'loop gain' of the circuit

It can be expressed as  $\beta.A_v$

**20. Define open loop gain?**

The gain obtained from the circuit without considering the feedback network is known as open loop gain.

**21. Define closed loop gain?**

The gain obtained from the oscillator circuit considering the feedback network is known as closed loop gain.

**22. List out the classification of oscillators?**

- i) Based on output waveform
  - a) Sinusoidal
  - b) Non-sinusoidal
- ii) Based on circuit component
  - a) RC oscillators
  - b) LC oscillators
- iii) Based on range of operating frequency
  - a) Audio frequency oscillators
  - b) Radio frequency oscillators
- iv) Based on feed back
  - a) Feedback oscillator
  - b) Non-feedback oscillators

**23. Define RC oscillator.**

The oscillator using the components like resistors (R) and capacitor (C) to generate low (or) audio frequency signals are called RC oscillators (or) audio-frequency oscillators.

Example: RC-phase shift oscillators, Wein bridge oscillator

**24. Define LC oscillators.**

The oscillators using a tuned circuit consisting of components like inductance (L) and capacitance (C) to generate high frequency signals are called LC oscillators (or) radio-frequency oscillators.

**Example:** Hartley, Colpitts, Clapp oscillator, etc..

**25. What is Barkhausen criterion?**

The condition required to work the circuit as an oscillator is given by

- “The total phase shift around a loop, as the signal proceeds from input through amplifier, feedback network back to input again, computing a loop is precisely  $0^\circ$  (or)  $360^\circ$ ”



- “The magnitude of the product of the open loop gain of the amplifier ( $A_v$ ) and the magnitude of the feedback factor ( $\beta$ ) is unity. (i.e.)  $|A_v\beta|=1$ .”

## 26. Define Frequency stability of an oscillator?

The ability of an oscillator to maintain a constant frequency of oscillation is called frequency stability. The oscillation frequency depends on various factors like stray capacitance, temperature, etc.

## 27. List the factors that affect the frequency stability.

- Operating point of active devices
- Dependency of Circuit components on temperature.
- Supply voltage.
- Output load.
- Inter-element capacitances & stray capacitances.

## 28. What are the advantages the wein bridge oscillator?

- It gives constant output.
- The circuit works quite easily
- The overall gain is high because of two resistance
- The frequency of oscillators can be easily changed by using potentiometer.

## 29. List out the disadvantages of Wein bridge oscillator?

- The circuit requires two transistors and a large number of components.
- It cannot generate very high frequencies of oscillations.

## 30. What are the types of resonating frequency?

- Series resonating frequency
- Parallel resonating frequency.

## 31. Difference between series and parallel resonance frequency?

| Sl.no. | Series resonance frequency                                                                                                | Parallel resonance frequency                                                                                                  |
|--------|---------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| 1.     | Series resonance frequency occurs when the reactance of the inductance (L) is equal the reactance of capacitance( $C_1$ ) | Parallel resonance frequency occurs when the reactance of R-L- $C_1$ branch is equal to the reactance of capacitance( $C_2$ ) |
| 2.     | Impedance is minimum                                                                                                      | Impedance is maximum                                                                                                          |

|    |                                                                     |                                                                                                           |
|----|---------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| 3. | Series resonance frequency is given by<br>$f_s = 1/2\pi\sqrt{LC_1}$ | Parallel resonance frequency is given by<br>$f_p = 1/2\pi\sqrt{LC_{eq}}$ $C_{eq} = (C_1.C_2)/(C_1 + C_2)$ |
|----|---------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|

**SRI MANAKULA VINAYAGAR ENGINEERING COLLEGE**

(Approved by AICTE, New Delhi & Affiliated to Pondicherry University)  
(Accredited by NBA-AICTE, New Delhi, ISO 9001:2000 Certified Institution &  
Accredited by NAAC with "A" Grade)

Madagadipet, Puducherry - 605 107

**UNIT-V****1. What is an op-amp?**

An operational amplifier is a DC-coupled high-gain electronic voltage amplifier with a differential input and, usually, a single-ended output.[1] An op-amp produces an output voltage that is typically hundreds of thousands times larger than the voltage difference between its input terminals.

**2. What are the different linear IC packages?**

There are three popular packages available

1. The metal can(TO) packages
2. The dual-in-line packages(DIP)
3. The flat package of flat pack

Op-amp packages may contain single, two (dual) or four (quad) op-amps. Typical packages have 8 terminals, 10 terminals and 14 terminals.

**3. What are the basic terminals of the OP-AMP?**

There OP-AMP has five basic terminals .They are

- Two input terminals
  - Inverting terminal (Pin No 2)
  - Non Inverting Terminal (Pin No 3)
- One output terminal (Pin No 6)
- Two Power Supply Terminals(Pin No 4 for -Vcc and 7 for +Vcc)

**4. What are the DC characteristics of an OP-AMP?**

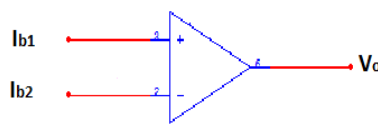
The non-ideal dc characteristics that add error components to the dc output voltage are:

- Input bias current
- Input offset current
- Input offset voltage

- Thermal drift

### 5. What is input-bias current?

The average amount of current entering into the Inverting and the Non inverting terminals of the op-amp is called Input bias current. Its value is 500nA for IC741A



The input-bias current( $I_b$ ) is given by

$$I_b = (I_{b1} + I_{b2}) / 2$$

### 6. What is the input-offset current?

Even before applying the inputs to the inverting and the Non inverting terminals there is a small amount of output is obtained because the terminals are not electrically balanced. In order to rectify this we apply the input offset current.

The algebraic difference between the currents into the Inverting and the Non inverting input terminals to make the output as zero is referred to as input offset current. It is 200nA maximum for IC741C

### 7. What is the input-offset voltage?

Even before applying the inputs to the inverting and the Non inverting terminals there is a small amount of output is obtained because the terminals are not electrically balanced. In order to rectify this we apply the input offset voltage.

The voltage that is applied between the inverting and the Non-inverting terminals of the OP-AMP to make the output as zero. Hence we nullify the output .So it is called as Input Offset Voltage.

### 8. What is thermal drift?

The DC characteristics like the Bias current, offset voltage change with temperature. The change of the values of the DC characteristics with temperature is called as Thermal Drift. Offset current Drift is expressed in nA/ $^{\circ}$ C and offset voltage drift in mV/ $^{\circ}$ C.

This effect can be reduced by Using Proper printed circuit board layout and by Forced air cooling.

**9. What are the AC characteristics of an OP-AMP?**

The AC characteristics of an OP-Amp are

- Frequency Response
- Frequency Compensation
- Slew Rate
- Supply Voltage Rejection Ratio
- Large Signal Voltage Ratio
- Gain Bandwidth Product
- Transient Response

**10. What is slew rate?**

It is defined as the maximum ratio of change of output Voltage per unit Time. It is always expressed in terms of Volts per micro seconds. For example, a 1V/us slew rate means that the output rises or falls by 1V in one micro seconds.

An ideal slew rate is infinite which means that op-amp's output voltage should change instantaneously in response to input step voltage.

**11. What is input resistance and input capacitance in an OP-AMP?**Input Resistance:

It is defined as the resistance measured between any one of the input terminal to ground by making the other input terminal already connected to the ground

Input Capacitance:

It is defined as the capacitance measured between any one of the input terminal to ground by making the other input terminal already connected to the ground.

**12. Explain the method for increasing the input resistance of an Op-Amp?**

1. One of the methods to obtain the higher values of Input Resistance is by using Darlington Pair in the place of transistors.

2. Another method to increase the Input Resistance is to fabricate a FET differential pair as the input stage with the rest of the stages made by BJT. The Input Resistance of the order of  $10^{12}$  ohms is possible with such JFET inputs.

**13. What is the input voltage range for OP-AMP?**

This voltage is the common mode voltage that can be applied to both input terminals without disturbing the performance of the op-amp. For the 741C the range of the input common-mode voltage is (+or -) 13V. Common mode configuration is used only for test purpose to determine the degree of matching between the inverting and the non inverting terminals.

**14. What is CMRR?**

The relative sensitivity of an op-amp to a difference signal as compared to a common mode signal is called as common mode rejection ratio (CMRR) and gives the figure of merit  $\rho$  for the differential amplifier. So CMRR is given by:

$$\rho = \left| \frac{A_{dm}}{A_{cm}} \right|$$

And it is expressed in decibel (dB). For example the uA741 op-amp has a maximum CMRR of 70 dB.

**15. What is SVRR? (supply voltage rejection ratio).**

The change in an OP-AMP's input offset voltage due to variations in supply voltage is called the supply voltage rejection ratio. Some manufactures use the terms like Power Supply Rejection ratio (PSRR) or Power supply sensitivity (PSS). These terms are expressed in micro volts per volts or in decibel For IC741c, SVRR-150uV/V. Lower the value of the SVRR, better the op-amp.

**16. What is large signal voltage gain?**

An op-amp amplifies the difference voltages between the two input terminals and therefore, its voltage gain is defined as, "The ratio between the output voltage to the differential input voltage"

$$\text{Voltage Gain} = \frac{\text{output voltage}}{\text{differential input voltage}}$$

Since the amplitude of the output signal is much larger than the input signal, the voltage gain is referred as large signal voltage gain.

**17. What is output voltage swing?**

The output voltage swing indicates the value of positive and negative saturation voltages of an op-amp and never exceeds the supply voltage. For IC741C, the output voltage swing is guaranteed to be between +13V and -13V for  $R_L > 2\text{kohms}$ .

**18. What is Output Resistance?**

The output Resistance  $R_o$  is the resistance measured between the output terminal of the op-amp and the ground. The Output Resistance for the IC 741C op-amp is 75ohms.

**19. What is Transient Response?**

The rise time and overshoot are the two characteristics of the transient response of any circuit. These parameters are of importance whenever selecting an op-amp for ac applications. The transient response test circuit is included in the data sheet for IC741C, Rise time is 0.3us and overshoot is 5%.

**20. What is Power-supply rejection ratio?**

The output of a perfect operational amplifier will be completely independent from ripples that arrive on its power supply inputs. Every real operational amplifier has a specified power supply rejection ratio (PSRR) that reflects how well the op-amp can reject changes in its supply voltage.

**21. State the various methods of improving CMRR?**

The methods of improve CMRR are,

- a. Constant current bias method
- b. Use of current mirror circuit
- c. Use of active load

**22. What are the applications of comparator?**

1. Zero crossing detectors
2. Window detector
3. Time marker generator

**23. What is a switching regulator?**

Switching regulators are those which operate the power transistor as a high frequency on/off switch, so that the power transistor does not conduct current continuously. This gives improved efficiency over series regulators.

**24. What is the advantage of switching regulators?**

Greater efficiency is achieved as the power transistor is made to operate as low impedance switch. Power transmitted across the transistor is in discrete pulses rather than as a steady current flow. By using suitable switching loss reduction technique, the switching frequency can be increased so as to reduce the size and weight of the inductors and capacitors.

**25. What are the applications of Op-Amp?**

- Inverting amplifier
- Non-inverting Amplifier
- Integrator
- Comparator
- Subtractor
- Summer
- Analog Multiplier